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Alzheimer's Disease Research



Mahdi Moghaddami

Projects Overview

Longitudinal vs. Cross-Sectional

- Longitudinal Models:
  - RNN
  - LSTM
  - GRU
  - MinimalRNN
- Cross-Sectional Models:
  - Random Forest
  - Support Vector Machine
  - Logistic Regression

- This study compares two categories of modeling: longitudinal and cross-sectional. We predict the clinical diagnosis of cognitive impairment in Alzheimer's disease (CN, MCI, or AD).

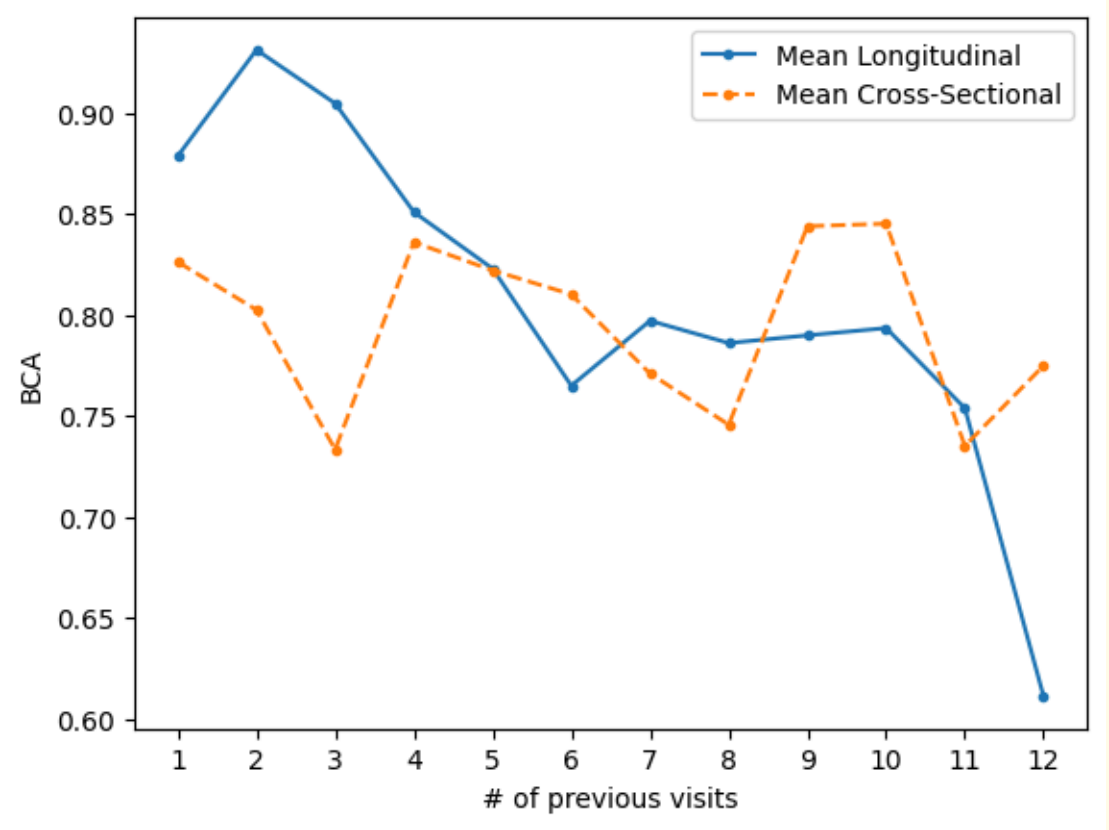


Figure 1: Mean BCA for top 3 models in each category

Review Article

- Two major questions:
  1. How is longitudinal data used for AD?
  2. How should we write articles when there is an abundance of papers?
     - ASReview
     - Elicit
     - Covidence
     - DistillerSR
     - Nested-Knowledge

- 2020-2026 papers
- Four Sources:
  1. ACM Digital Library
  2. IEEEXplore
  3. Scopus
  4. PubMed
- We filter papers based on the existence of certain keywords in the title and abstract
- We find relevant papers.
- We need further ranking of paper. We add different metric for papers:
  - Journal papers:
    - Journal Impact Factor (JIF)
    - CiteScore
    - SCImage Journal Rank (SJR)
    - Source Normalized Impact per Paper (SNIP)
  - Conference papers:
    - Core Ranking (A, A, B, C)
    - CCF Ranking (A, B, C)
  - Both:
    - Google Scholar h5-index

Longitudinal vs. Cross-Sectional

- The question:
  - Will a subject progress to a further stage of AD (CN → MCI or MCI → AD) in the next visit?
- Binary Stable/Converter prediction.

- 337 subjects
- The data:
  - DTI
  - Structured clinical and biomarker data (TADPOLE)
- Brett will talk more about this project!

AD Progression Prediction using fMRI

What is fMRI?

- Functional Magnetic Resonance Imaging is a type of brain scan that shows both the structure of the brain and, more importantly, **which areas are active during specific tasks or at rest**.
- It’s acquired by tracking blood flow changes. When a brain area is more active, it uses more oxygen, and fMRI can detect that via something called the **BOLD signal** (Blood Oxygen Level Dependent signal).
- Each voxel contains a BOLD signal that changes over time.
- It’s basically **a number of 3D images**.



Figure 2: Example fMRI volume

Preprocessing Steps using fMRIPrep

1. **Convert the data into a standard format:**
   - DICOM (.dcm) → NIfTI (.nii)
2. **Motion correction:**
   - During a several-minute fMRI scan, the subject inevitably moves a little.
   - fMRIPrep estimates how each volume moved relative to a reference volume.
   - It then realigns the volumes.
3. **Slice-timing correction:**
   - An fMRI volume isn't actually acquired all at once.
   - The scanner acquires different slices at slightly different times.
   - Slice-timing correction estimates what the signal would have been at the appropriate timing.
4. **Susceptibility distortion correction:**
   - Magnetic fields get warped near air-filled spaces in the skull (like the sinuses).
   - This step calculates that magnetic warping and "un-squishes" the image back to its true anatomical shape.
5. **Coregistration:**
   - fMRI images are low-resolution and blurry.
   - To know exactly where the brain activity is happening, fMRIPrep aligns the functional image with the subject's anatomical image (T1w MRI).
6. **Spatial normalization:**
   - Every person's brain is a slightly different size and shape.
   - fMRIPrep estimates a nonlinear transformation that maps the individual's anatomy to a standard brain template (usually called MNI space).

Generating Brain Functional Connectivity Network using NiLearn

1. **Parcellation (applying an atlas):**
   - Instead of analyzing 100,000 tiny individual voxels, Nilearn uses a brain "atlas" to group those voxels into larger, meaningful regions of interest (ROIs) (e.g., the hippocampus, the amygdala, the motor cortex).
   - $R = \{R_1, R_2, ..., R_N\}$: set of ROIs defined by the atlas
2. **Extracting time series from each ROI (4D → $|R| \times T$):**
   - Each voxel in an ROI has its own BOLD time series. We summarize these voxel signals, typically by taking the mean signal across voxels in the region.
   - $R_1 = [x_1, x_2, ..., x_T]$ where T is the number of time points.
   - We do this for every region.
3. **Correlation matrix ($|R| \times T \rightarrow |R| \times |R|$):**
   - "How similarly does Region i's activity fluctuate with Region j's activity?"
   - Suppose:
     - $R_i = [0.2, 0.4, 0.1, -0.2, ...]$
     - $R_j = [0.3, 0.5, 0.2, -0.1, ...]$
   - We calculate the Pearson correlation:

$$r_{ij} = \text{corr}\left(R_i, R_j\right)$$

$$= \frac{\sum(R_i - \mu_i)(R_j - \mu_j)}{\sqrt{\sum(R_i - \mu_i)^2}\sqrt{\sum(R_j - \mu_j)^2}}$$

   - $-1 \leq r_{ij} \leq 1$ for all $i, j \in \{1, ..., N\}$

FC Matrix

$$FC = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1N} \\ r_{21} & r_{22} & \cdots & r_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ r_{N1} & r_{N2} & \cdots & r_{NN} \end{bmatrix}$$

- An example functional connectivity (FC) matrix using the AAL atlas (number of regions: 116):

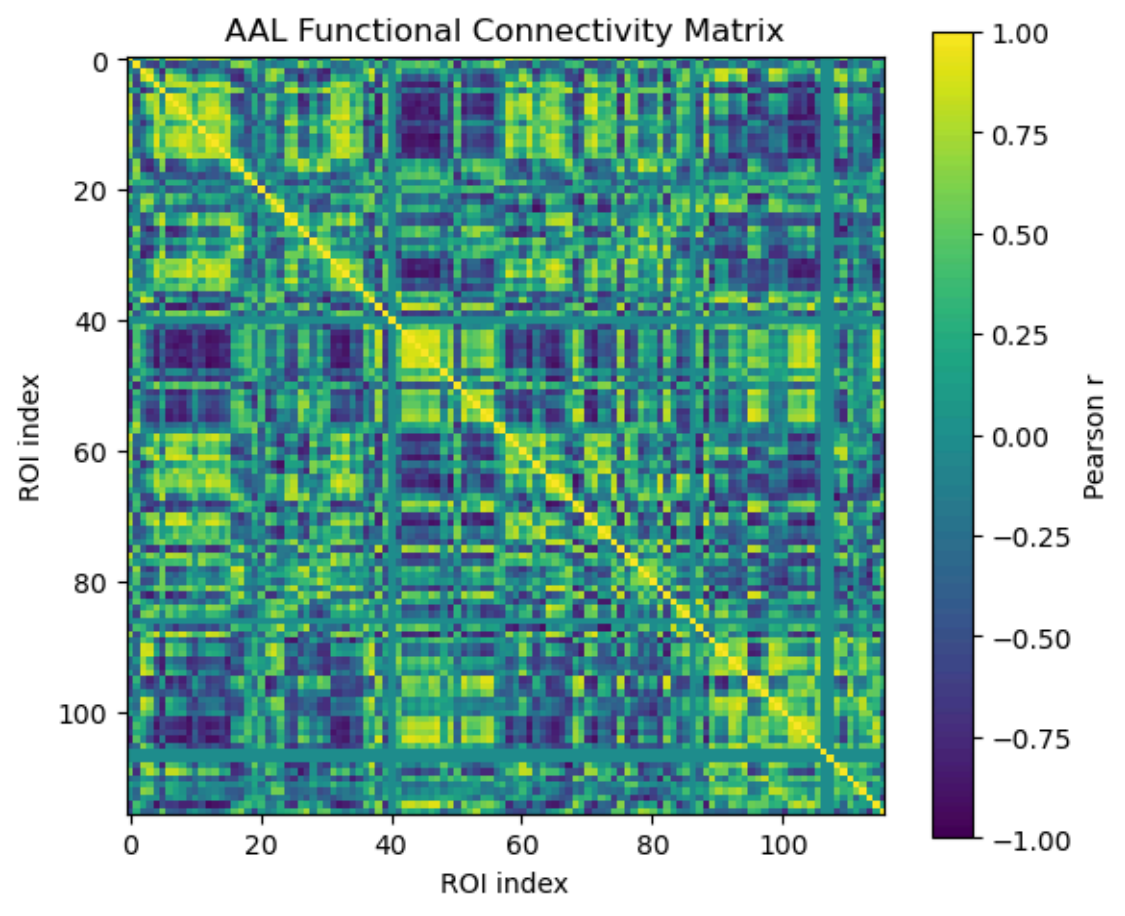


Figure 3: Example FC matrix

The Dataset

- **Stable sequence:** Last visit diagnosis same as previous
- **Converter sequence:** Last visit diagnosis different from previous (CN → MCI or MCI → AD)

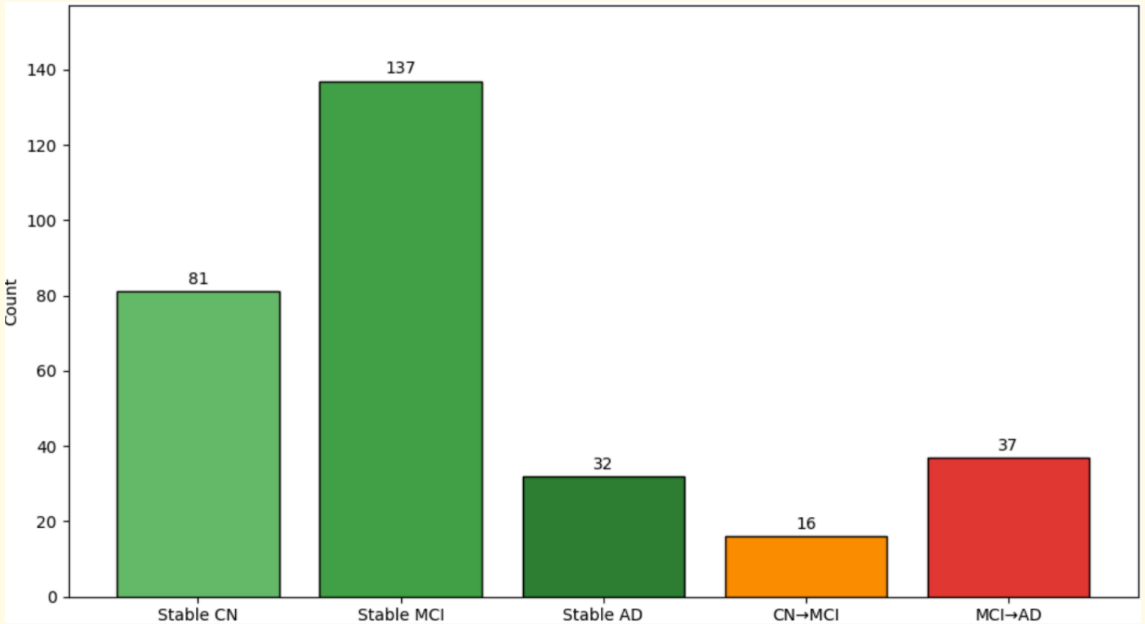


Figure 4: Stable/converter subject distribution

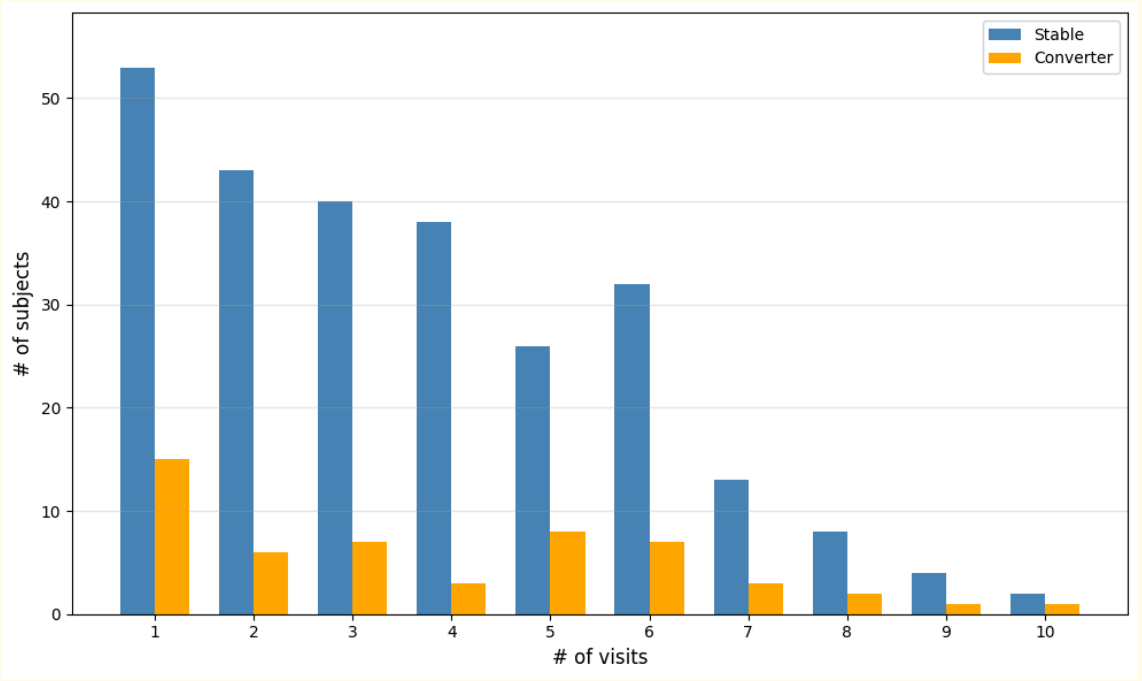


Figure 5: Another view of stable/converter subject distribution

The Model

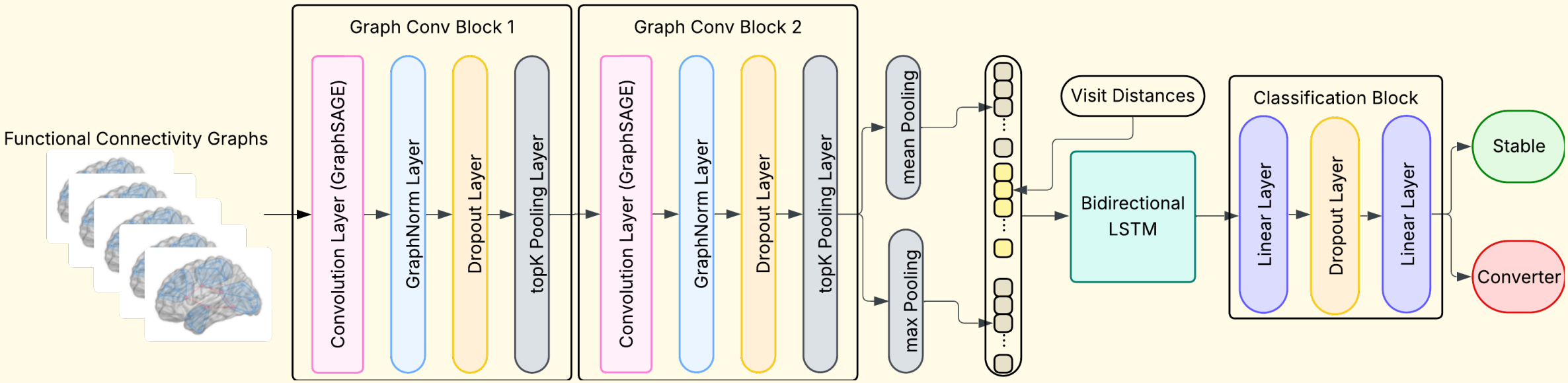
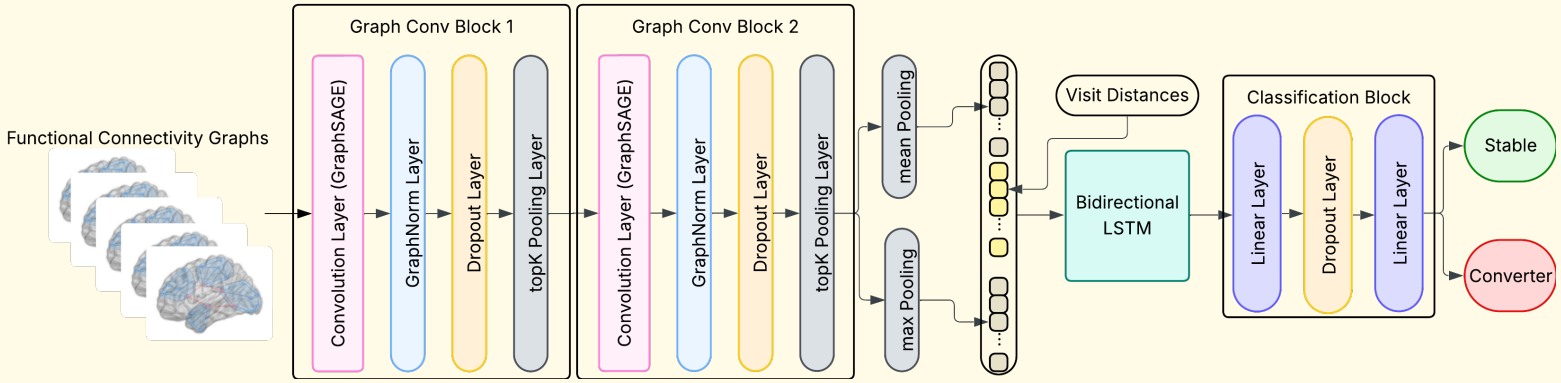


Figure 6: Model architecture

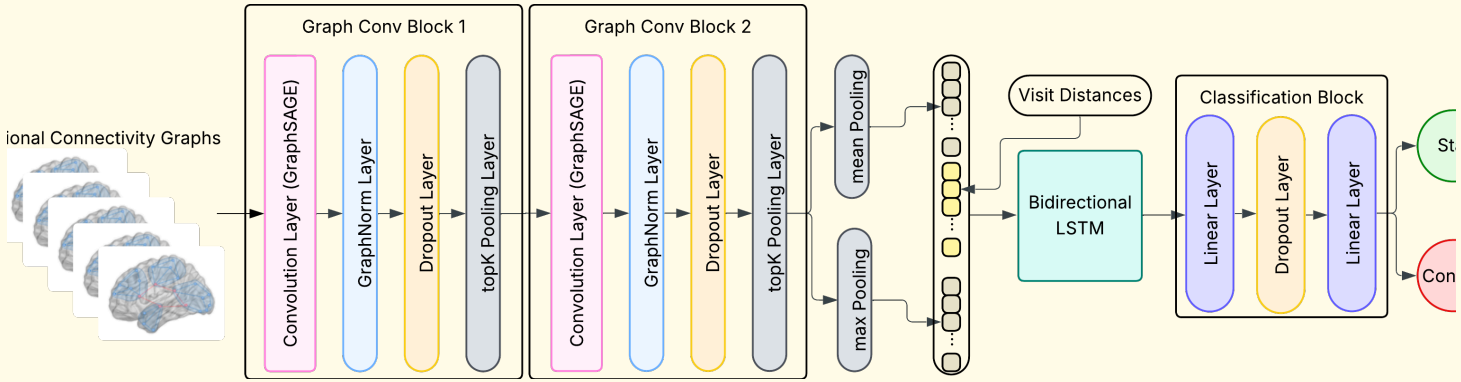
Input



- The original 2D functional connectivity matrix is $N \times N$, where N is the number of ROIs.
- For the GNN input, that matrix is converted into a graph:
  - Nodes = V = ROIs
  - Edges = E = functional connections between ROIs
    - An edge exists only if the correlation between two nodes is $\geq$ *threshold*. We set *threshold*=0.2.
  - Node features = connectivity profile of each ROI (its row of the FC matrix)

hello

GraphSAGE [NeurIPS 2017]



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GraphSAGE [NeurIPS 2017]

Connectivity profile of ROI i

$$x_i = [\mathrm{corr}(i, R_1), ..., \mathrm{corr}(i, R_N)]$$

Mean of neighboring node features

$$m_i = \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} x_j$$

Updated node embedding

$$h_i = \sigma\left(W_{\text{self}} x_i + W_{\text{neigh}} m_i + b\right)$$

Notation

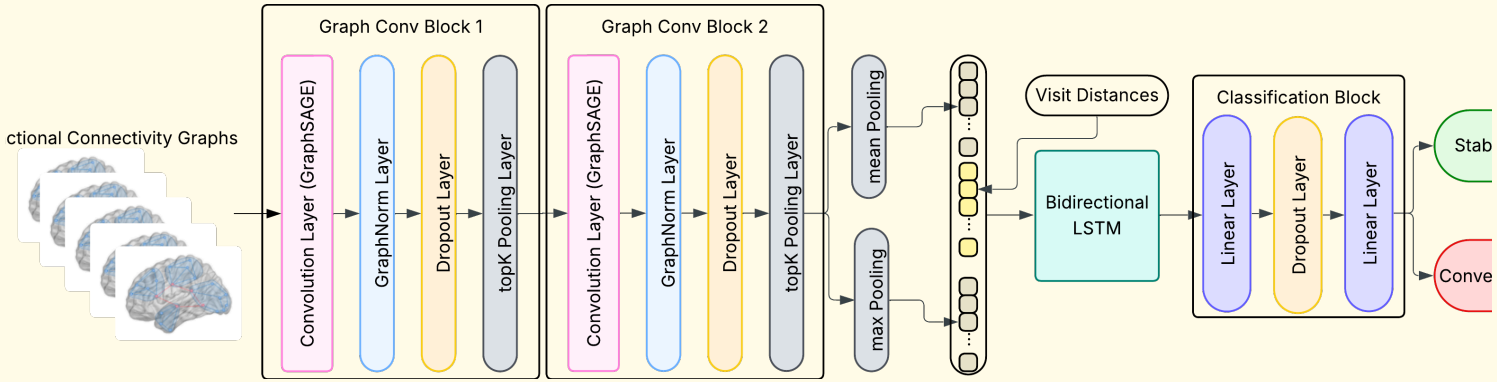
$\mathcal{N}(i)$: neighbors of ROI i
$W_{\text{self}}, W_{\text{neigh}}$: learned weights
b: learned bias
σ: activation function

Layer output

$$H = \begin{pmatrix} h_1^T \\ \vdots \\ h_N^T \end{pmatrix} \in \mathbb{R}^{N \times d}$$

N nodes, d embedding features

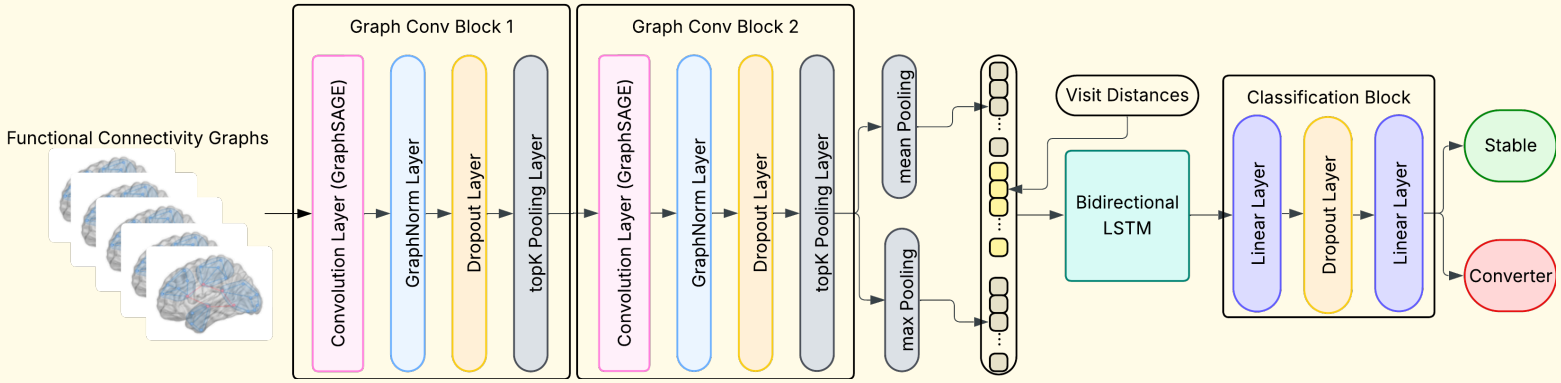
GraphNorm [ICML 2021]



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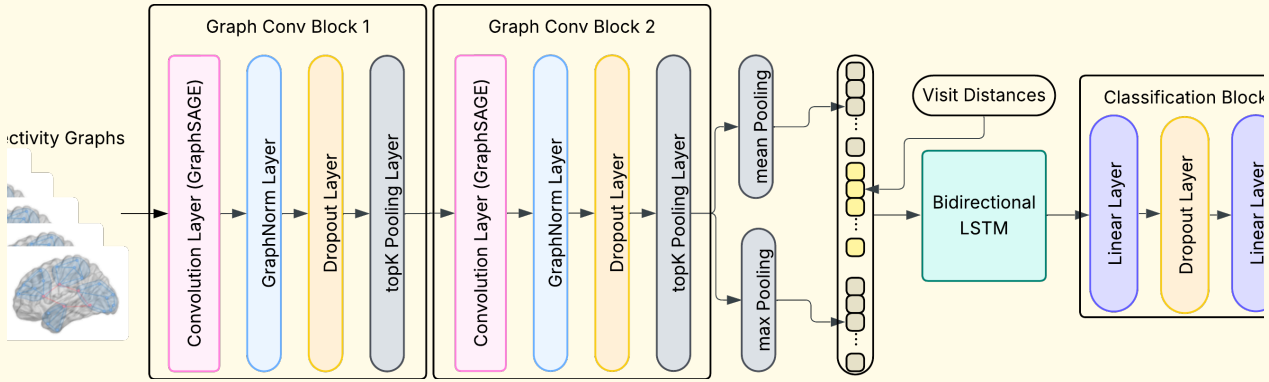
Dropout [JMLR 2014]



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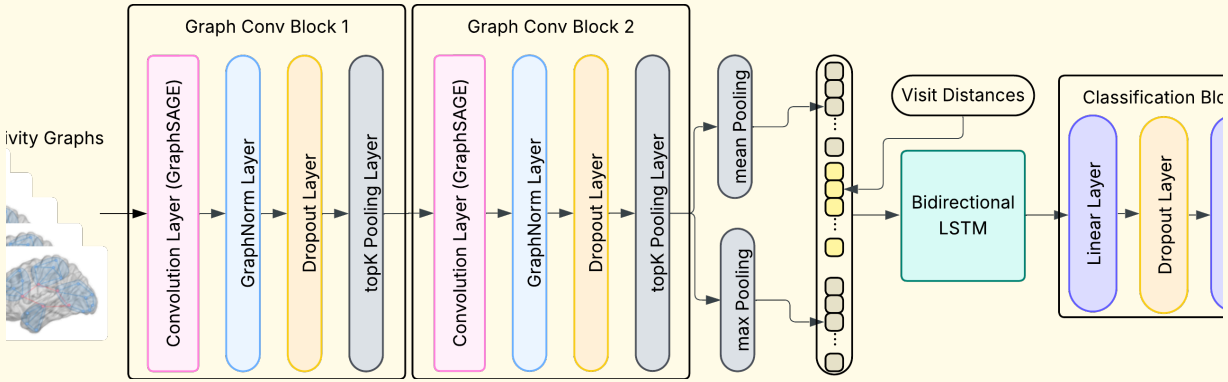
TopK Pooling [RRL@NeurIPS 2018]



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Mean and Max Pooling



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Pretraining

- Using 20% of the data

Challenges

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Thank you!